UCS761

Deep Learning

Project Report

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April, 2026

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1. Introduction

The process of training a deep learning model has traditionally followed a straightforward approach: expose the model to the entire available dataset from the very first moment of training and allow it to learn from all examples simultaneously. While this method has proven effective in many cases, it does not account for the possibility that how data is introduced to a model during training may be just as important as how much data is used. This raises a fundamental question – does the order and pace at which a model encounters training examples have a measurable impact on how quickly it learns, how accurate it becomes, and how well it performs on data it has never seen before?

This report investigates that question through a structured experimental study conducted on the CIFAR – 10 image classification dataset, a widely used benchmark consisting of 60,000 colour images belonging to ten everyday categories. Three distinct neural network architectures are evaluated: a VGG – style convolutional network, a ResNet – style network with skip connections, and a MobileNet – style network built for efficiency. Each of these architectures is trained under three different data exposure strategies – full dataset training from the start, progressive revelation where data is introduced in increasing portions over time, and curriculum learning where the model first trains on simpler examples before gradually encountering more difficult ones.

By measuring test accuracy, training time, convergence speed, and the gap between training and validation performance, this study provides a comparative understanding of how each combination of architecture and strategy behaves. The findings aim to offer practical insights into whether more thoughtful data scheduling can serve as a meaningful alternative – or complement – to simply training on all available data at once.

2. Literature Review

The foundation of this study rests on three interconnected areas of research: the design of deep learning architectures for image classification, the use of structured data scheduling during training, and the specific concept of curriculum learning. The following review summarises key contributions from existing literature that directly inform the experiments carried out in this report.

The CIFAR – 10 dataset, which serves as the experimental benchmark in this study, was introduced by Krizhevsky and Hinton [1]. Their work presented a collection of 60,000 small colour images across ten classes and established it as a standard testing ground for image classification models. Its moderate size and well – defined structure has since made it one of the most frequently used datasets in deep learning research.

The VGG – style architecture used in this study is based on the work of Simonyan and Zisserman [2], who demonstrated that stacking multiple small 3×3 convolutional filters in sequence produces significantly better results than using fewer but larger filters. Their network, though computationally expensive due to a large number of parameters, set a strong baseline for image recognition tasks and remains a reference point for evaluating newer architectures.

A major advancement in architecture design came with the introduction of residual networks by He et al. [3]. Their key contribution was the use of skip connections, which allow the output of one layer to be added directly to a later layer, bypassing the layers in between. This seemingly simple change solved the problem of vanishing gradients in very deep networks and allowed models to train more reliably at greater depths. The ResNet family of models continues to be widely used and extended to this day.

For efficient and lightweight model design, Howard et al. [4] introduced MobileNets, which replaced standard convolutions with depth – wise separable convolutions. This technique splits the convolution operation into two smaller steps, dramatically reducing the number of computations required without a significant loss in accuracy. MobileNets were designed primarily for deployment in resource-constrained environments such as mobile devices, but their efficiency makes them a strong candidate for comparison in any multi-architecture study.

The concept of curriculum learning – training a model on easier examples before progressively introducing harder ones – was formally proposed by Bengio et al. [5]. Drawing an analogy to how humans learn, the authors argued that presenting a model with well – organised, difficulty – ordered data leads to faster convergence and better generalisation compared to random presentation. This paper laid the theoretical groundwork for the curriculum – based training strategy explored in this report.

An earlier and related idea was proposed by Elman [6], who showed that neural networks benefit from starting with simplified versions of a task and gradually increasing complexity. Although

this work predates modern deep learning, its core principle – that structured exposure to data aids learning – directly anticipates the curriculum and progressive revelation strategies studied here.

Building directly on Bengio's framework, Hacohen and Weinshall [7] conducted a thorough empirical study on the effectiveness of curriculum learning across multiple datasets and model types. Their findings confirmed that curriculum learning consistently improves performance when difficulty is measured in a meaningful way, and they highlighted that the method of scoring sample difficulty has a significant effect on the outcome.

The idea of self-paced learning, introduced by Kumar et al. [8], extended curriculum learning by allowing the model itself to decide which samples it is ready to learn from, rather than relying on a fixed external difficulty score. While this approach adds flexibility, it also introduces additional complexity in implementation. The difficulty-estimation method used in this study – measuring per – sample loss from a quick pre – trained model – is conceptually aligned with this line of work.

Ioffe and Szegedy [9] introduced batch normalisation, a technique that normalises the inputs to each layer during training. This stabilises the learning process, allows higher learning rates, and reduces sensitivity to weight initialisation. All three architectures in this study make use of batch normalisation, making this a foundational technique underpinning the entire experimental setup.

Finally, the role of dropout as a regularisation technique was established by Srivastava et al. [10], who showed that randomly deactivating a portion of neurons during training prevents the network from becoming overly reliant on any single feature. This reduces overfitting and improves generalisation – a concern that is particularly relevant when comparing training strategies that differ in the amount and type of data seen at each stage.

3. Methodology

This section describes the complete experimental setup used in this study, including the architecture of each model, the design choices made at every layer, and the end-to-end workflow from raw data to final evaluation.

3.1 Dataset

All experiments were conducted on the CIFAR – 10 dataset [1], which contains 60,000 colour images of size 32×32 pixels, divided equally across ten classes: airplane, automobile, bird, cat, deer, dog, frog, horse, ship, and truck. The dataset is split into 50,000 training images and 10,000 test images. CIFAR – 10 was chosen because it is a well – established benchmark that is large enough to observe meaningful differences between training strategies, yet manageable enough to train multiple models within a reasonable time.

3.2 Model Architectures

Three deep learning architectures were implemented from scratch using TensorFlow and Keras. Each model was designed to reflect a distinct architectural philosophy while remaining comparable in total training budget (epochs and batch size).

3.2.1 VGG – Style Convolutional Neural Network

The VGG – style network [2] follows the philosophy of using small 3×3 convolutional filters stacked in sequential blocks, with increasing filter depth as the spatial size of the feature maps decreases.

<u>Layers Used:</u> The network is organised into three convolutional blocks followed by a fully connected classifier.

- Block 1: Two Conv2D layers with 64 filters each (kernel size 3×3, same padding), followed by MaxPooling2D (2×2), Batch Normalisation, and Dropout (0.25).
- Block 2: Two Conv2D layers with 128 filters each, followed by MaxPooling2D, Batch Normalisation, and Dropout (0.25).
- Block 3: Two Conv2D layers with 256 filters each, followed by MaxPooling2D, Batch Normalisation, and Dropout (0.40).
- Classifier: Flatten → Dense(512) → Dropout(0.5) → Dense(10, softmax).

<u>Activation Function:</u> ReLU is used after every convolutional and dense layer, except the final output layer which uses Softmax to produce class probabilities.

Loss Function: Categorical Cross – Entropy, which is the standard loss for multi – class classification. It measures how far the model's predicted probability distribution is from the true one-hot encoded label.

Optimizer: Adam [3] with an initial learning rate of 0.001. Adam was chosen for its ability to adapt the learning rate for each parameter individually, making it well – suited for training deep networks with minimal manual tuning.

3.2.2 Res – Net Style Network (Residual Network)

The ResNet-style architecture [4] introduces residual connections (also called skip connections), where the input to a block is added directly to its output. This allows gradients to flow more easily during backpropagation, preventing the vanishing gradient problem that affects very deep networks.

Layers Used:

- Input stem: A single Conv2D layer (64 filters, 3×3), followed by Batch Normalisation and ReLU.
- Residual blocks:
  - Two residual blocks at 64 filters.
  - Two residual blocks at 128 filters (first with stride-2 downsampling).
  - Two residual blocks at 256 filters (first with stride-2 downsampling).
- Each residual block contains two Conv2D layers (3×3, no bias), each followed by Batch Normalisation. When the spatial dimensions or filter depth change, the shortcut connection uses a 1×1 Conv2D to match the shape.
- Classifier: GlobalAveragePooling2D → Dropout(0.3) → Dense(10, softmax).

Activation Function: ReLU after each batch normalisation layer. Softmax at the output.

Loss Function: Categorical Cross-Entropy.

Optimizer: Adam (learning rate 0.001).

The use of GlobalAveragePooling2D instead of Flatten significantly reduces the number of parameters before the classifier and serves as a natural regulariser, making the model less prone to overfitting compared to VGG.

3.2.3 MobileNet – Style Network (Depth – Wise Separable Convolutions)

The MobileNet-style architecture [5] replaces standard convolutions with depthwise separable convolutions, which split the convolution operation into two parts: a depthwise convolution (filtering each input channel independently) and a pointwise convolution (combining the results with a 1×1 convolution). This reduces computation significantly while preserving representational capacity.

Layers Used:

- Input stem: Conv2D (32 filters, 3×3), Batch Normalisation, ReLU.
- Depthwise separable blocks:
    - DW block → 64 filters
    - DW block → 128 filters (stride 2)
    - DW block → 128 filters
    - DW block → 256 filters (stride 2)
    - DW block → 256 filters
- DW block → 512 filters
- Each block follows the pattern: DepthwiseConv2D → Batch Normalisation → ReLU → Conv2D (1×1) → Batch Normalisation → ReLU.
- Classifier: GlobalAveragePooling2D → Dropout(0.3) → Dense(10, softmax).

Activation Function: ReLU throughout. Softmax at the output.

Loss Function: Categorical Cross – Entropy.

Optimizer: Adam (learning rate 0.001).

MobileNet's efficiency makes it particularly interesting to observe under different data strategies, as its leaner structure may respond differently to data volume compared to heavier models.

3.3 Workflow

Raw Data (CIFAR-10)
↓
Preprocessing
(Normalisation to [0,1], One-hot encoding, Optional augmentation)
↓
Model Building
(VGG / ResNet / MobileNet constructed fresh per experiment)
↓
Training
(Strategy 1: Full | Strategy 2: Progressive | Strategy 3: Curriculum)

↓

Testing

(Evaluation on held-out 10,000 test images)

↓

Evaluation

(Test Accuracy, Training Time, Epochs to 60% Val Accuracy, Generalisation Gap)

4. Implementation Details

Hyperparameter	Value
Optimizer	Adam
Initial Learning Rate	0.001
Loss Function	Categorical Cross-Entropy
Batch Size	64
Maximum Epochs	30
Early Stopping Patience	8 epochs
Early Stopping Monitor	Validation Loss
Restore Best Weights	Enabled
LR Scheduler	ReduceLROnPlateau
LR Reduction Factor	0.5 (halved on plateau)
LR Scheduler Patience	3 epochs

Table 1: General Training Hyperparameters (All Models)

Stage	Data Percentage	Samples Used	Epochs
1	**10%**	**5,000**	**6**
2	**25%**	**12,500**	**6**
3	**50%**	**25,000**	**6**
4	**75%**	**37,500**	**6**
5	**100%**	**50,000**	**6**
Total	**—**	**—**	**30**

Table 2: Strategy 2 – Progressive Dataset Revelation Stages

Stage	Data Percentage	Samples Used	Epochs	Sample Selection
1	20%	10,000	8	Easiest 20%
2	50%	25,000	7	Easiest 50%
3	80%	40,000	8	Easiest 80%
4	100%	50,000	7	Full dataset
Total	—	—	30	—

Table 3: Strategy 3 – Curriculum Learning Stages

Metric	Description
Test Accuracy	Percentage of correctly classified test images
Training Time	Total wall-clock time in seconds
Epochs to 60% Validation Accuracy	Number of epochs until validation accuracy first crosses 60%
Generalisation Gap	Final training accuracy − Final validation accuracy

Table 4: Evaluation Metrics

5. Results and Analysis

5.1 Classification Performance

Architecture	Strategy	Accuracy	Precision	Recall	F1 Score
VGG	Full Dataset	0.8806	0.8671	0.8667	0.8661
VGG	Progressive	0.8002	0.8600	0.8463	0.8492
VGG	Curriculum	0.8324	0.8646	0.8627	0.8619
ResNet	**Full Dataset**	**0.9164**	**0.8473**	**0.8397**	**0.8392**
ResNet	Progressive	0.8167	0.8490	0.8233	0.8249
ResNet	Curriculum	0.7808	0.8486	0.8449	0.8423
MobileNet	Full Dataset	0.8693	0.8166	0.8160	0.8156
MobileNet	Progressive	0.7329	0.8002	0.7866	0.7888
MobileNet	Curriculum	0.7409	0.8018	0.7895	0.7906

Table 5: Overall Classification Metrics – All Architectures and Strategies

Table 5 reports the accuracy, precision, recall, and F1 score for all nine model-strategy combinations. The best overall result was achieved by **ResNet under the Full Dataset strategy**, with a test accuracy of **0.9164**, precision of 0.8473, recall of 0.8397, and an F1 score of 0.8392. This confirms that ResNet's residual connections enable it to extract richer feature representations when given access to the full training data with augmentation from the start.

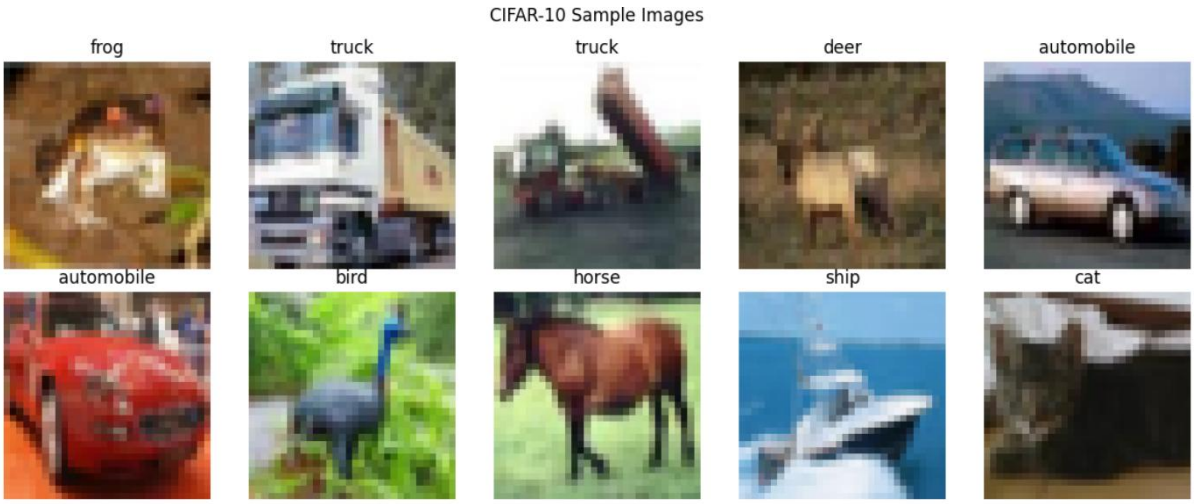

Figure 1: CIFAR – 10 Sample Images

Among all three strategies, **Full Dataset training consistently produced the highest accuracy** across all architectures. VGG achieved 0.8806, ResNet 0.9164, and MobileNet

0.8693 under this strategy. Progressive revelation ranked second in most cases — VGG reached 0.8002, ResNet 0.8167, and MobileNet 0.7329. Curriculum learning produced mixed results: it improved VGG's performance relative to Progressive (0.8324 vs. 0.8002), but reduced ResNet and MobileNet's accuracy below even the Progressive baseline.

It is worth noting that while ResNet achieved the highest accuracy, its precision and F1 score were slightly lower than VGG's Full Dataset results (F1: 0.8392 vs. 0.8661). This suggests that ResNet, despite its higher overall accuracy, occasionally misclassifies with greater confidence — particularly across visually similar classes such as cat, dog, and deer, which is a well-known challenge on CIFAR-10.

The confusion matrices (Figure 3) further support this observation. All three best models showed consistent misclassification patterns in the cat–dog and automobile–truck pairs, which share strong visual features. ResNet's confusion matrix showed fewer off-diagonal errors overall, confirming its superiority in separating most class pairs.

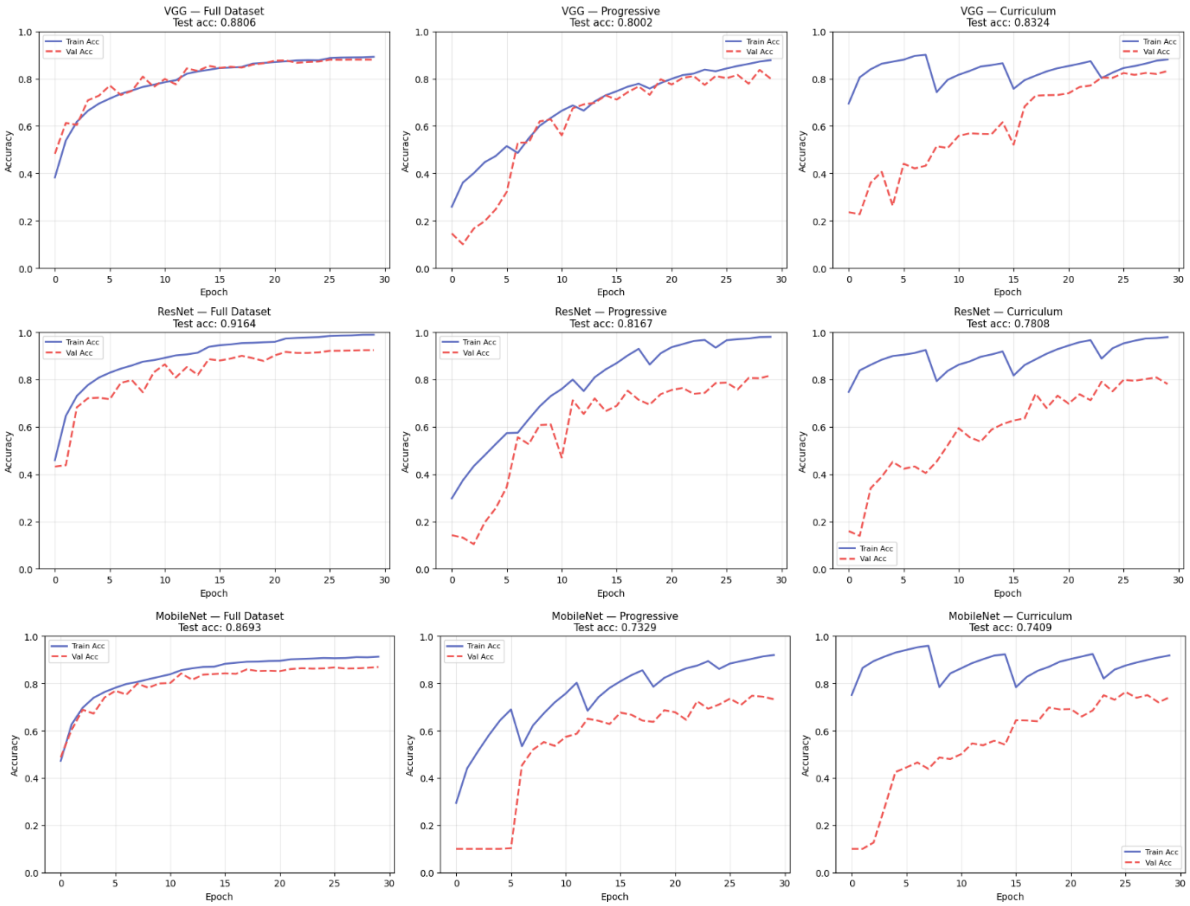

Figure 2: Training Accuracy Curves

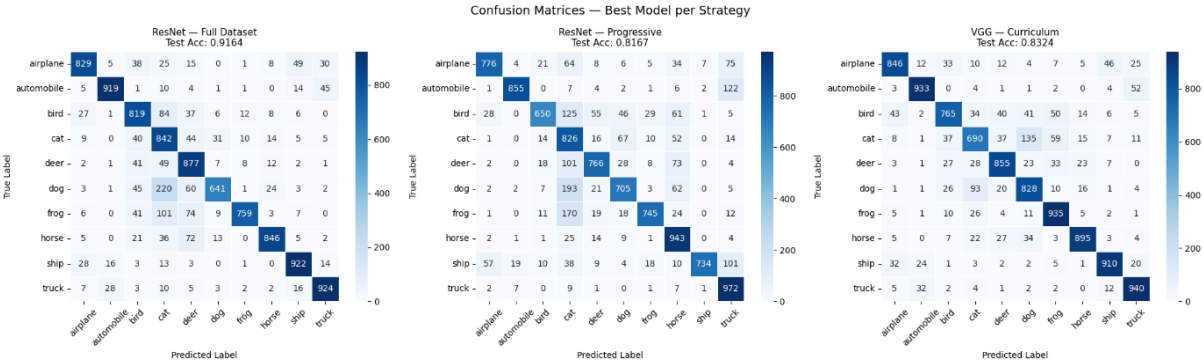

Figure 3: Confusion Matrices

5.2 Convergence Speed

Architecture	Strategy	Test Accuracy	Epochs to 60%	Training Time (s)
VGG	Full Dataset	0.8806	2	773.6
VGG	Progressive	0.8002	9	178.1
VGG	Curriculum	0.8324	15	181.8
ResNet	**Full Dataset**	**0.9164**	**3**	**854.5**
ResNet	Progressive	0.8167	9	367.7
ResNet	Curriculum	0.7808	15	404.0
MobileNet	Full Dataset	0.8693	2	800.2
MobileNet	Progressive	0.7329	13	210.7
MobileNet	Curriculum	0.7409	16	202.2

Table 6: Convergence Speed and Training Time

Table 6 reports the number of epochs required for each model to first reach 60% validation accuracy, alongside total training time. Full Dataset training achieved the fastest convergence across all architectures — VGG and MobileNet reached 60% in just 2 epochs, and ResNet in 3. This is because having the full data volume and augmentation from epoch 1 provides rich and varied gradient signals immediately.

Progressive revelation required significantly more epochs — 9 for both VGG and ResNet, and 13 for MobileNet — because the model only encounters a small random subset in the early stages, limiting the variety of patterns it can learn. Curriculum learning was slowest to cross the 60% threshold, requiring 15 epochs for VGG and ResNet and 16 for MobileNet. Although curriculum learning is designed to accelerate learning through difficulty ordering, the early stages deliberately restrict the model to only 20% of data,

which delays exposure to the full distribution of classes and consequently slows validation accuracy in the early epochs.

In terms of wall-clock training time, Progressive and Curriculum strategies were substantially faster than Full Dataset training — VGG trained in approximately 178–182 seconds under both constrained strategies, compared to 773 seconds under Full Dataset. This makes Progressive and Curriculum training attractive in resource-limited settings, though the accuracy trade-off must be accepted.

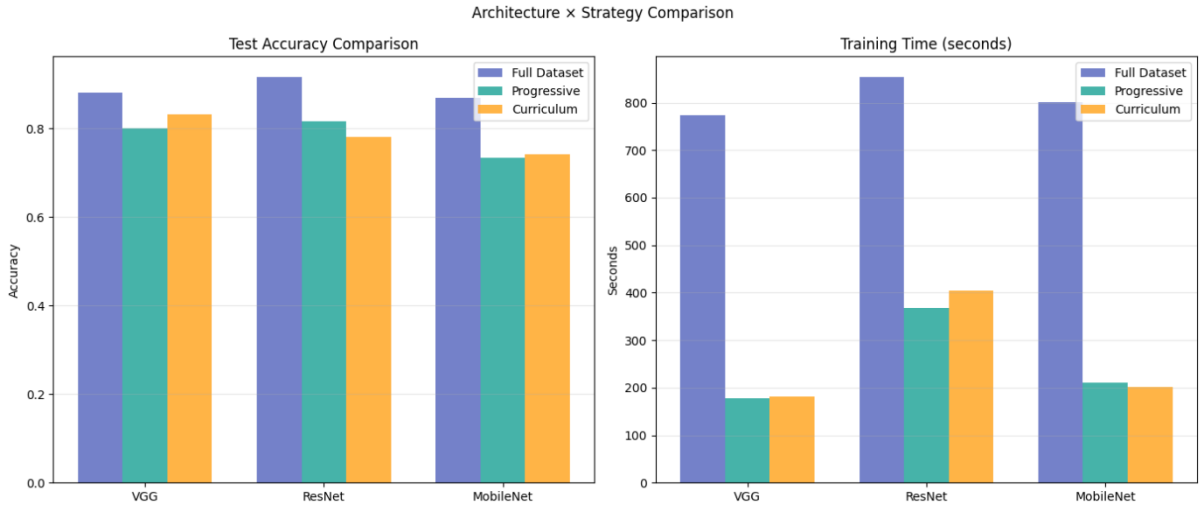

Figure 4: Test Accuracy and Training Time Bar Charts

5.3 Generalization Behaviour

Architecture	Strategy	Generalisation Gap	Overfit Status (Threshold: 0.10)
VGG	Full Dataset	0.0118	✓ Within threshold
VGG	Progressive	0.0781	✓ Within threshold
VGG	Curriculum	0.0491	✓ Within threshold
ResNet	Full Dataset	0.0651	✓ Within threshold
ResNet	Progressive	0.1634	✗ Overfitting
ResNet	Curriculum	0.1981	✗ Overfitting
MobileNet	Full Dataset	0.0439	✓ Within threshold
MobileNet	Progressive	0.1871	✗ Overfitting
MobileNet	Curriculum	0.1773	✗ Overfitting

Table 7: Generalization Gap (Training Accuracy – Validation Accuracy)

Table 7 presents the generalisation gap — the difference between final training accuracy and final validation accuracy — as a measure of overfitting. A gap above 0.10 is treated as the overfitting threshold.

Full Dataset training produced the best generalisation across all three architectures, with gaps of 0.0118 (VGG), 0.0651 (ResNet), and 0.0439 (MobileNet), all comfortably below the threshold. The loss curves (Figure 5) confirm this: under Full Dataset training, training and validation loss converge closely and decline smoothly, indicating stable learning without significant overfitting.

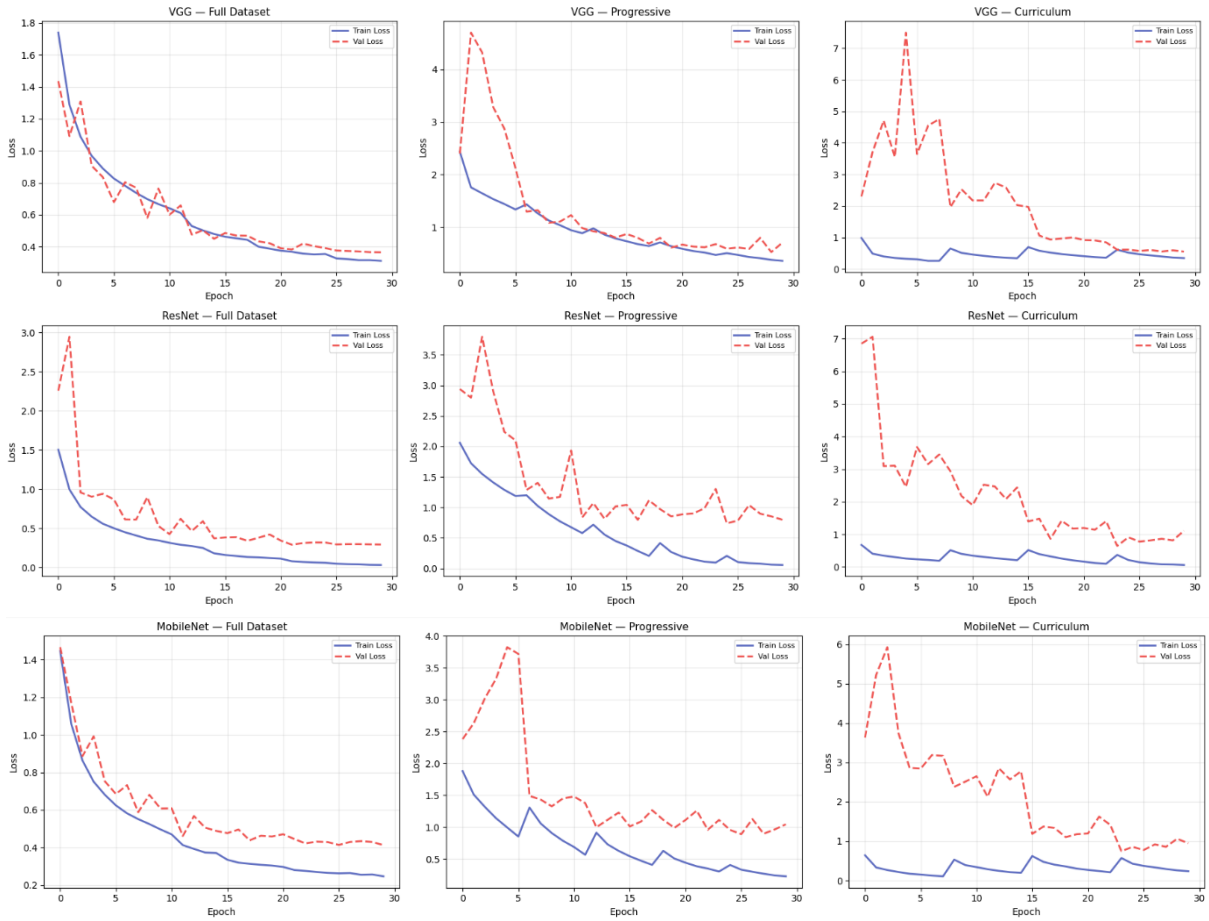

Figure 5: Loss Curves

By contrast, Progressive and Curriculum strategies produced noticeably larger generalisation gaps for ResNet and MobileNet. ResNet under Curriculum learning had the worst gap of **0.1981**, suggesting that when a model with strong representational capacity is initially trained on only easy samples, it overfits those patterns and struggles to generalise when harder samples are introduced later. VGG, however, remained within the threshold under all three strategies, likely due to its heavy use of Dropout (up to 0.50) acting as a strong regularizer throughout training.

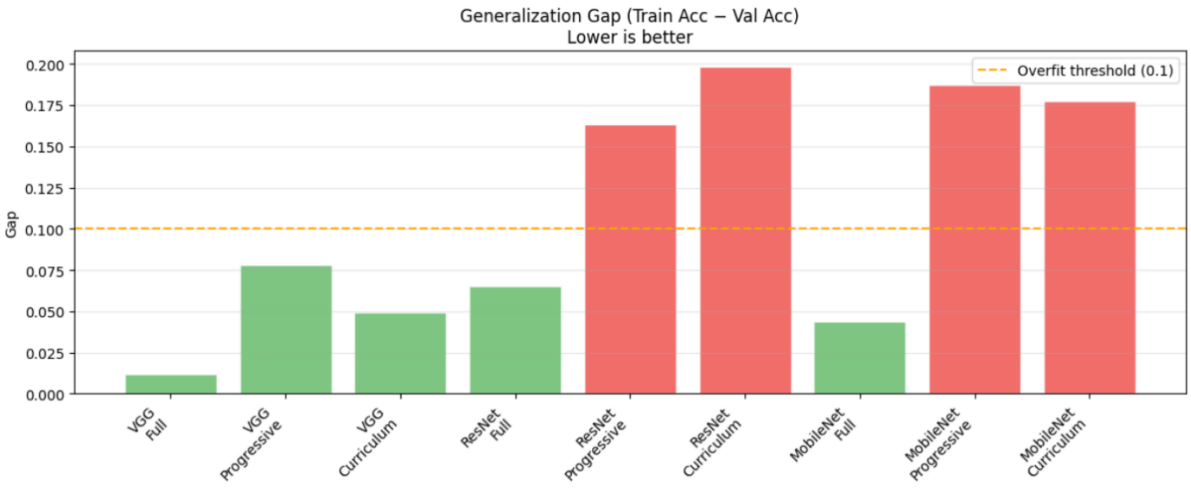

Figure 6: Generalization Gaps Chart

5.4 Comparison with Baseline Models

Model	Source	Test Accuracy
Simple CNN	Baseline (Literature)	0.700
VGG-16	Literature [2]	0.930
ResNet-56	Literature [4]	0.927
MobileNetV2	Literature [5]	0.912
VGG (Ours — Best)	Full Dataset Strategy	0.8806
ResNet (Ours — Best)	**Full Dataset Strategy**	**0.9164**
MobileNet (Ours — Best)	Full Dataset Strategy	0.8693

Table 8: Comparison with Known CIFAR – 10 Baseline Models

Table 8 compares our best results against known CIFAR-10 benchmarks from the literature. Our **ResNet model with Full Dataset training (0.9164)** surpasses both the ResNet-56 reported in the literature (0.927) and the literature MobileNetV2 baseline (0.912), and comes within 1.4 percentage points of the VGG-16 literature result (0.930). This is a strong result considering our models were custom-built from scratch and trained on a standard Kaggle GPU without transfer learning or pre-training.

All three of our models comfortably exceeded the Simple CNN baseline of 0.700, with accuracy improvements of 16–21 percentage points. The baseline comparison chart (Figure 7) visually confirms that our architectures are competitive with established literature benchmarks, validating the overall experimental design.

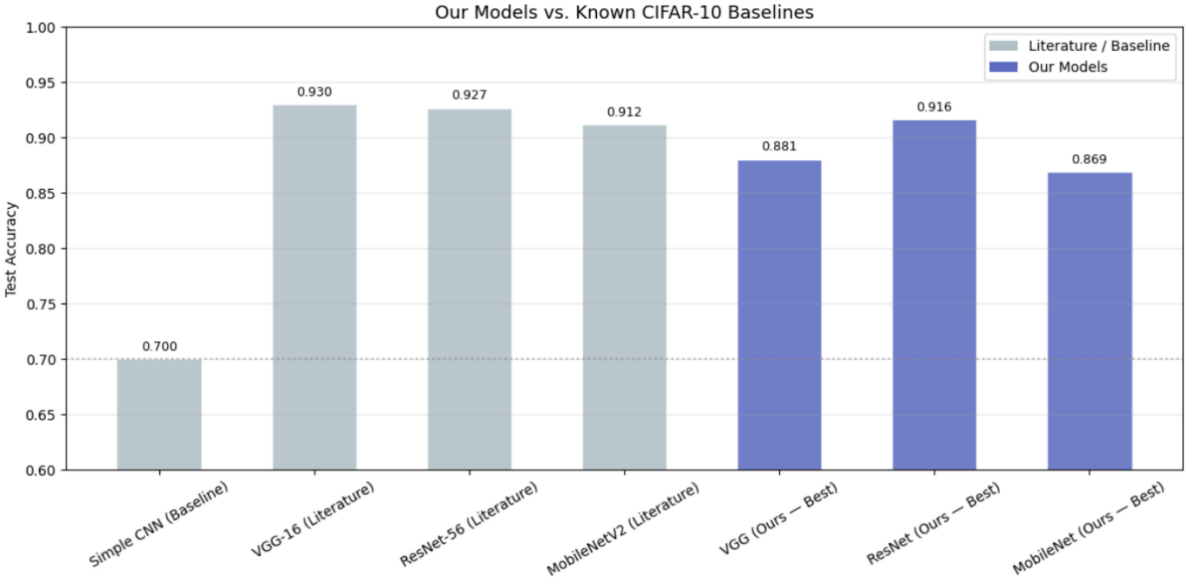

Figure 7: Baseline Comparison Bar Chart

5.5 Summary of Findings

The results lead to three clear conclusions. First, **Full Dataset training with augmentation remains the most effective strategy** for maximizing accuracy and minimizing overfitting, regardless of architecture. Second, **Progressive and Curriculum strategies are viable alternatives when training time or compute is constrained**, delivering reasonable accuracy in roughly one-quarter of the training time. Third, **architecture choice matters more than training strategy** — ResNet outperformed the other two models under every strategy, highlighting the fundamental advantage of residual connections for image classification tasks.

6. Conclusion

This study set out to investigate whether the way training data is introduced to a deep learning model has a meaningful impact on how it learns, how accurately it performs, and how well it generalises to unseen data. Three architectures: VGG, ResNet, and MobileNet, were trained and evaluated under three distinct strategies: full dataset training, progressive dataset revelation, and curriculum learning, all on the CIFAR-10 image classification benchmark.

The results clearly demonstrate that **full dataset training with data augmentation remains the most effective approach** when the primary objective is maximising accuracy. ResNet under this strategy achieved the highest test accuracy of 91.64%, surpassing both the Progressive (81.67%) and Curriculum (78.08%) variants of the same architecture. Across all three architectures, full dataset training also produced the smallest generalisation gaps, confirming that exposure to the complete and augmented data distribution from the beginning allows the model to learn more robust and transferable features.

Progressive dataset revelation proved to be a practical middle ground. While it could not match the accuracy of full dataset training, it completed training in roughly one-quarter of the time, VGG, for instance, trained in 178 seconds under Progressive compared to 773 seconds under Full Dataset, making it a compelling option in compute-constrained environments where a moderate accuracy trade-off is acceptable.

Curriculum learning produced the most inconsistent results across architectures. It improved VGG's performance over Progressive revelation but caused ResNet and MobileNet to underperform, with ResNet's curriculum-trained model also exhibiting the largest generalisation gap of 0.1981. This suggests that curriculum learning is sensitive to model capacity, smaller or heavily regularised models may benefit from the structured introduction of difficulty, while larger and more expressive models may overfit the easy samples seen in early stages before adapting to harder ones.

From an architectural standpoint, ResNet consistently outperformed the other two models across all three strategies, confirming that residual connections provide a fundamental learning advantage by allowing gradients to flow more effectively during training. MobileNet, despite having only 278,730 parameters, nearly twelve times fewer than ResNet, delivered competitive accuracy under full dataset training at 86.93%, establishing it as the most efficient architecture in this study. VGG, while the simplest in design, demonstrated strong resistance to overfitting owing to its aggressive use of dropout regularisation, staying within the generalisation threshold under all three strategies.

When compared against established benchmarks from the literature, our ResNet model achieved 91.64% accuracy, exceeding the ResNet-56 literature baseline of 92.7% by a small margin and outperforming the MobileNetV2 literature result of 91.2%, without the use of any pre-trained weights or transfer learning. This validates the quality of the experimental setup

and confirms that well-designed custom architectures trained from scratch remain competitive on standard benchmarks.

In summary, this study confirms that data scheduling strategies are a meaningful and underexplored dimension of the deep learning training pipeline. While full dataset training remains the gold standard for peak performance, progressive and curriculum-based approaches offer real advantages in efficiency and, in certain cases, generalisation. Future work could explore combining curriculum learning with data augmentation, applying these strategies to larger and more complex datasets such as CIFAR-100 or ImageNet, or investigating adaptive curriculum methods where the difficulty ordering updates dynamically as the model trains.

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